PGR111, solutions week 08

Task 1 - Concepts & Terminology

Briefly explain the following terminology and database concepts:

- Surrogate key
- Cardinality (for relationships)
- Participation (for relationships)
- Multiplicity (for relationships)
- Super key
- Candidate key

Note: Keep your answers. They might become helpful later in the semester. :-)

Look in the slides, the textbook or on the internet for answers to this one.

Task 2 – ER Modell & SQL

Feel free collaborate with others on the tasks (but you can do the tasks alone too).
Note: We recommend using paper or **Lucidchart/Gliffy/etc.**, rather than the "data modelling" part of MySQL Workbench: the latter skips (automates) some parts of the tasks, thus you might miss out on some of the intended learning.

Imagine that you have been hired to design a database solution for a university college (like Kristiania). They have the following thoughts on the content. **Create an ER model for part 1-5.**

1. They will store student data in the database, so they easily can send out information, etc. Each student should have a unique student number, as well as the possibility to store a name, an address, and several emails (e.g.: school email, private email, ...).

   Task 1-5 has its own solution pdf: DB1102-PGR111_08_Solutions_ER-Models.

2. The courses at the school should be stored with an id (6 characters, both numbers and letters) and a title (100 characters).

   Task 1-5 has its own solution pdf: DB1102-PGR111_08_Solutions_ER-Models.

3. Which field of study each student has chosen must also be registered (each student can only follow one field of study).

   Task 1-5 has its own solution pdf: DB1102-PGR111_08_Solutions_ER-Models.

4. Each field of study must be linked to the courses it contains. Although it is common to have courses per field of study, it should also be possible to have common courses for several fields of study.

   Task 1-5 has its own solution pdf: DB1102-PGR111_08_Solutions_ER-Models.

5. Complete the ER model with the details needed for a working database (add PKs + FKs and replace any M:M relationships with composite tables).

   Task 1-5 has its own solution pdf: DB1102-PGR111_08_Solutions_ER-Models.

6. Implement the tables in MySQL! (Create a new database/schema: college).

```
create table FieldOfStudy
(
    id int not null auto_increment,
    name varchar(100) not null,
    constraint fieldOfStudy_id_pk primary key (id)
);

create table Student
(
    studNr int not null auto_increment,
    name varchar(100) not null,
    address varchar(100),
    idFieldOfStudy int not null,
    constraint student_studnr_pk primary key (studNr),
    constraint student_idFieldOfStudy_fk foreign key (idFieldOfStudy)
        references FieldOfStudy(id)
);

create table Email
(
    eMail varchar(100) not null,
    studNr int not null,
    constraint email_email_pk primary key (email),
    constraint email_studnr_fk foreign key (studNr)
        references Student(studNr)
);

create table Course
(
    id char(6) not null,
    title varchar(100),
    constraint course_id_pk primary key (id)
);
```

```
create table Courses_ForEach_FieldOfStudy
(
    idFieldOfStudy int not null,
    idCourse char(6) not null,
    constraint courses_forEach_fieldOfStudy_idFieldOfStudy_idCourse_pk
        primary key (idFieldOfStudy, idCourse ),
    constraint courses_forEach_fieldOfStudy_idFieldofStudy_fk
        foreign key (idFieldOfStudy) references FieldOfStudy(id),
    constraint courses_forEach_fieldOfStudy_idCourse_fk
        foreign key (idCourse) references Course(id)
);
```

7. Register the following courses - id / title: *(feel free to add more courses yourself!*

   DS1337 / Cool design
   PG1011 / Binary programming
   PJ1234 / Basic project

```
insert into course
values ('DS1337', 'Cool design'),
       ('PG1011', 'Binary programming'),
       ('PJ1234', 'Basic project');
```

8. Register 2 fields of study: Dynamic Design and Pragmatic Programming.
   Make connections between fields of study and courses so that the result is:
   - Pragmatic Programming contains the courses PJ1234 and PG1011.
   - Dynamic Design contains the courses PJ1234 and DS1337.

```
insert into FieldOfStudy
values (NULL, 'Dynamic Design'),
       (NULL, 'Pragmatic Programming');

insert into Courses_ForEach_FieldOfStudy
values (2, 'PJ1234'),
       (2, 'PG1011'),
       (1, 'PJ1234'),
       (1, 'DS1337');
```

9. Register yourself as a student in one of the field of studies.

```
insert into student
values (NULL, '<your name>', '<your adrdess>', <id FieldOfStudy>);
```

10. Create a query to find out what courses you're following based on your name.

    There are several possible solutions here:

```
-- With JOIN, version 1:
select c.title
from Course c
join Courses_ForEach_FieldOfStudy cf on c.id = cf.idCourse
join FieldOFStudy f on cf.idFieldOfStudy = f.id
join student s on f.id = s.idFieldOfStudy
where s.name = '<your name>';

-- JOIN, version 2: We can optimize it – we can skip fieldOfStudy, and connect
-- student directly with Courses_ForEach_FieldOfStudy, since both student and
-- Courses_ForEach_FieldOfStudy has a FK to FieldOfStudy:
select c.title
from course c
join Courses_ForEach_FieldOfStudy eff on c.id = eff.idCourse
join student s on eff. idFieldOfStudy = s.idFieldOfStudy
where s.name = '<your name>';

-- We can also solve it with subqueries:
select title
from course
where id in
  (select idCourse
   from Courses_ForEach_FieldOfStudy
   where idFieldOfStudy in
     (select idFieldOfStudy
      from student
      where name = '<your name>'
     )
  );
```

Task 3 – Modelling task from UCF

As an extra Modelling task set, solve this one from the University of Central Florida:
https://www.cs.ucf.edu/courses/cop4710/fall2013/hw1KEY%20-%20new.pdf
(Solutions are provided in the pdf, just after each question.)

It includes usage of today's extra ER diagram symbols (named "participation"), and the first task is a "reverse diagram"-thingy where you see the diagram and then explain what it means. 😊

Task 4 – Even more modelling

This is the end of the modelling topic & exercises. But it will be a part of the exam!

So, either:

- Look online for more modelling tasks that you can solve on your own.

- Or think about what domain you want to work on for your exam hand-in (a favourite hobby, or the like) and start planning an ER model for that domain.

Can’t help you with a solution on this part!! :-P